Review of run-20261015T1657Z: what the preregistered verdicts do and do not show

Kind: solver-axis stage diagnostic (AGENTS.md §2). Interpretation of the committed rows of `runs/*.jsonl` against `PREREGISTRATION.md` (§1–§5, Amendment 1). It changes no row, verdict or threshold. The run report is `reports/run-20261015T1657Z/REPORT.pdf`. Nothing here prices a pipeline, reports S or shows a speedup.

Verdicts as preregistered

Every series read **candidate** ($\sigma \le 0.25$ on ≥ 3 completed cells) except A0, N and A4 at m = 3 ($\sigma = 0.5$, **no middle ground**). Those labels stand as the protocol's output. The four observations below bear on what they mean.

1. At m = 2 the threshold does not discriminate

The incumbent A0 reads **candidate** at m = 2 ($\sigma = 0.19$, D = 4,4,4,4,5,5,5,5 over t = 2–9), and so does the null N. The $\sigma \le 0.25$ threshold came from the m = 4, n = 131 cost model (README §1, `BreakEven.lean`). At m = 2 every presentation measured, structured or not, has a slope under it. A **candidate** reading at m = 2 therefore carries no information about an arm.

2. The null equals the incumbent: D measures the presentation's shape

N draws a different curve, set and target in every cell (checked: `curve`, `x_r` and `planted_x` differ on all 26 row pairs) and gives different F4 traces (`steps` at m = 2, matrix sizes at m = 3). Its D equals A0′s in every cell at both arities. At these sizes D is set by the system's support and degrees, not by the curve. So differences between arms are differences between **presentations**, which is the question this study asks. No arm other than A0 has a null of its own shape, so none of them can be credited with structure beyond its shape.

3. D does not track cost for A1z and A1g

A1z and A1g sit at D = 8 in every completed cell. That is the input degree of their systems (the substituted x is quadratic, so S_3 has degree 8), hence $\sigma = 0$. Their matrices are the widest of all arms at every t, and their time grows fastest. At m = 2 and t = 6, the widest matrix has 42 652 columns against A0′s 2 451, and the time is 110 s against 0.3 s (target 0). Both series hit the memory limit first. For this arm σ on D is a saturated metric. The **candidate** label is the protocol's literal output, and these rows give no reason to expect it to replicate as a cheaper design. The preregistered prediction A1z = A0 ± 1 failed (+2 to +4). A1g = A1z ± 1 held (0 in every cell): dropping the smoothness condition costs nothing in D here.

4. A11 at m = 3: a constant-factor shift, not a flatter slope

At m = 3, A11 is the one arm below the incumbent at equal b on both D and matrix width.

m, t	A0 D	A11 D	A0 width	A11 width	A0 time	A11 time
3, 2	5	4	368	62	0.0 s	0.0 s
3, 3	6	4	3 285	781	0.6 s	0.1 s
3, 4	6	4	12 314	3 096	16.5 s	1.8 s
3, 5	≥ 6 (timeout)	≥ 5 (timeout)	≥ 97 456	≥ 40 486	300 s	300 s
2, 9	5	4	14 922	13 579	92.0 s	82.9 s

Its width grows by a factor of ×12.6 and then ×4.0 per bit at m = 3, against ×8.9 and ×3.7 for A0. Its growth rate is no flatter: the curves in the graph below run parallel, about 4× lower. At m = 2, A11′s width meets A0′s by b = 6–9, and both time out at t = 10. At m = 3 the t = 5 cell timed out at a partial D of 5, so A11′s D is not flat beyond the 3 completed cells. Its $\sigma = 0$ rests on the minimum number of cells the protocol accepts, on one prime, with no null of its own shape.

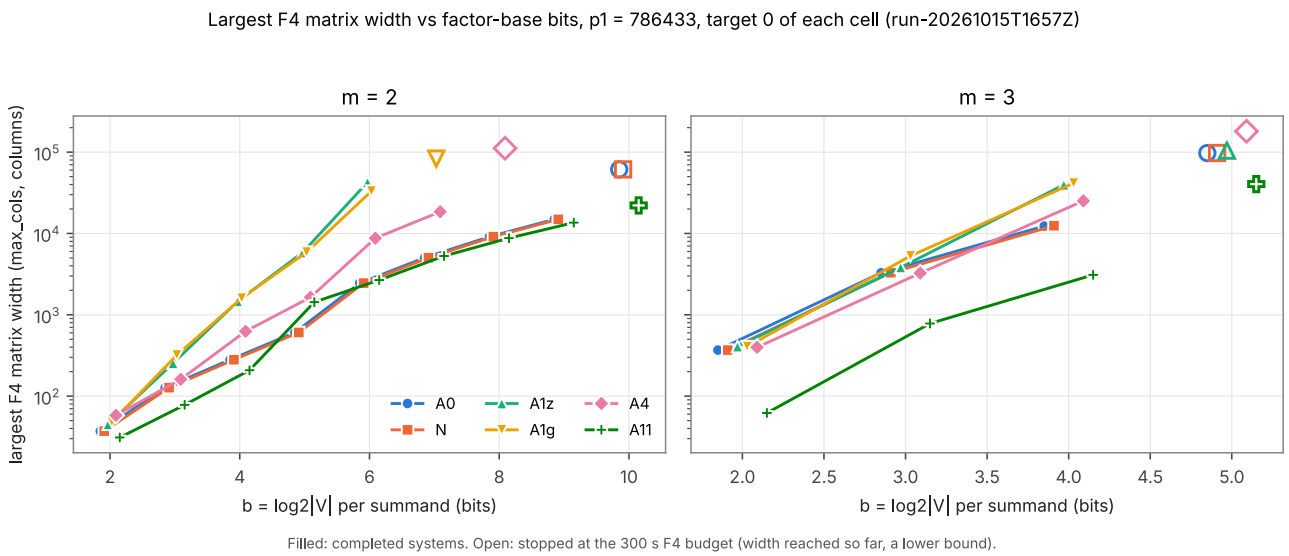


Figure 1: Largest F4 matrix width (`max_cols`, log scale) against b, per arm, target 0 of each cell. Open markers: systems stopped at the 300 s budget, at the width reached (a lower bound). Rows from `runs/*.jsonl`, plotted by `plot_cols.py`. The D graph is in the run report.

Predictions scorecard (§3)

prediction	observed	held?
A1z: D within ±1 of A0, σ no smaller	+2 to +4; σ = 0 (saturated at the input degree)	no
A1g = A1z ± 1	0 in every shared cell	yes
A4 = A0(⌈t/2⌉) + ≤ 1 (depth)	+1 to +2	no
A4 = A0(t) (Bezout)	+1 in 8 of 9 shared cells, 0 in one (m = 3, t = 3)	no; closer to Bezout + 1
A11: no prediction	lowest or tied-lowest D at every shared t (tied with A0 at m = 2, t = 5)	—

What is owed before any further statement

1. §5 replication of every **candidate** on p_2 = 2013265921. Only A11 at m = 3 is worth the cost, because points 1 and 3 empty the other labels.
2. A null of A11′s shape: the same norm-chain support with random coefficients and a planted zero. Without it, A11′s lower D cannot be told apart from the generic behaviour of a quadratic chain.
3. The A11 m = 3, t = 5 cell completed (a longer budget), giving a fourth point.
4. An m = 4 series, where the threshold was derived and A0′s slope is about 1 (`RESEARCH_DREG_MEASUREMENT.md`).

Each of these needs its own dated amendment before it runs.